Output Data Analysis for a Single System

Chapters 9.1 - 9.3 | Simulation Study Notes

9.1 Output Data Analysis: An Overview

When a simulation model is executed, it produces a sequence of output values that serve as the raw material for drawing conclusions about the system being modeled. Output data analysis refers to the rigorous statistical examination of these results to ensure that meaningful and accurate inferences can be drawn.

The Core Problem

A common and serious error in simulation practice is to perform only a single run of the model and treat the resulting output as a definitive answer. This approach is fundamentally flawed because simulation relies on random number generation, meaning that each execution of the model produces a different set of results. A single run may deviate substantially from the true long-run average of the system, and cannot be regarded as a reliable estimate.

Nature of Simulation Output

Simulation output is inherently random and subject to high variance. Each replication of the simulation yields different numerical values, and any one of these values may be far from the true performance measure of interest. This probabilistic character makes it essential to treat simulation as a statistical experiment conducted on a computer, applying the tools of probability and statistics to interpret the results correctly.

Why Standard Statistical Methods Are Insufficient

The output sequence (Y_1, Y_2, Y_3, ...) generated by a simulation is, in general, neither independent nor identically distributed. This means that standard statistical formulas that assume independence and identical distribution cannot be directly applied. In particular, simulation output is autocorrelated: successive observations are statistically dependent on one another. For example, the queue length observed at one point in time is influenced by the queue length at the previous instant, since customers who were waiting earlier continue to affect the system.

Replications and Independence

To overcome the problem of within-run dependence, simulation studies are typically conducted over multiple independent replications. If n replications are performed, each of length m, the output can be arranged as follows:

Run 1: $y_{11}, y_{12}, ..., y_{1m}$ Run 2: $y_{21}, y_{22}, ..., y_{2m}$ ... Run n: $y_{n1}, y_{n2}, ..., y_{nm}$

A fundamental property of this structure is that observations within the same replication (i.e., the same row) are dependent on one another, whereas corresponding observations from different replications (i.e., the same column across rows) are statistically independent. For instance, y_{11}, y_{21}, and y_{31} are independent, while y_{11} and y_{12} are not. This independence across replications is exploited to construct valid statistical estimates.

Goals of Output Analysis

The primary objectives of output data analysis are to estimate the true performance measures of the system, to construct confidence intervals that quantify the uncertainty of those estimates, and to measure the accuracy of the results. Two common difficulties arise in this process: the presence of a biased estimator, in which the estimated value systematically differs from the true value; and an incorrect variance estimate, which leads to errors in the stated precision of the results.

Illustrative Example: Bank System

Consider a bank with five tellers where customers arrive and are served according to random processes. Across ten independent simulation replications, the number of customers served ranged from 451 to 502, and the average customer delay ranged from 1.24 to 2.86 minutes. This wide variation across replications clearly illustrates why a single run cannot be considered representative of the system's true behavior. Only through the aggregation and statistical analysis of multiple runs can reliable conclusions be drawn.

9.2 Transient and Steady-State Behavior

The output sequence of a simulation (Y_1, Y_2, Y_3, ...) exhibits fundamentally different statistical behavior depending on how far the simulation has progressed from its starting point. Understanding this distinction is critical to the correct analysis of simulation output.

Transient Behavior

At the beginning of a simulation run, the output values are strongly influenced by the initial conditions chosen for the model. This early phase, in which the system has not yet settled into its natural pattern of operation, is referred to as the transient state. During this period, the statistical distribution of the output, denoted $F_i(y \mid I)$, changes with each time index i and depends on the initial condition I.

For example, if a queuing system begins with no customers present, the average delay will initially be very small, since the queue has not yet accumulated. As the simulation progresses, the queue builds up and the delay increases, causing the output to vary substantially over the early time periods. This instability is the defining characteristic of transient behavior.

Steady-State Behavior

After a sufficient period of time, the simulation reaches a condition in which the statistical properties of the output stabilize. This condition is known as the steady state. In steady state, the distribution of the output no longer changes with time; that is, Y_{k+1}, Y_{k+2}, ... all follow the same distribution for some time index k.

It is important to note that reaching steady state does not mean the output values themselves become constant. The individual observations continue to vary randomly. Rather, it is the underlying probability distribution that becomes stable. Furthermore, the observations in steady state are generally not independent of one another, as autocorrelation persists throughout the run.

Effect of Initial Conditions

While the steady-state distribution itself is independent of the initial conditions chosen for the simulation, the speed with which the system converges to steady state does depend on those conditions. A starting state that is closer to the typical steady-state behavior of the system will result in a faster convergence, thereby reducing the length of the transient period that must be discarded from the analysis.

Comparison: Transient vs. Steady-State

Aspect	Transient State	Steady State
Time Period	Early stage of simulation	Long-run stage of simulation
Distribution	Changes with time	Remains constant
Dependence on Initial Condition	Strong dependence	No dependence on final distribution
Output Values	Unstable, trending	Randomly varying around stable mean
Use in Analysis	Typically discarded (warm-up period)	Used for estimation

Example 9.2: M/M/1 Queue

In an M/M/1 queuing system, the delay experienced by the i-th customer, denoted D_i, serves as the output variable. As the simulation progresses, the expected value of D_i converges to a constant of 8.1. When the system begins with no customers (s = 0), the convergence is slower, whereas starting with 15 customers already in the system (s =

15) results in faster convergence. This confirms that initial conditions affect the rate of convergence to steady state but not the steady-state value itself.

Example 9.3: Inventory System

In an inventory simulation, the cost incurred in month i, denoted C_i, is observed over a long run. The expected cost converges to a steady-state value of E(C) = 112.11. Notably, the convergence is not monotone; the values fluctuate above and below the final mean before stabilizing. This non-smooth convergence is a common feature of simulation output and underscores the need to allow adequate warm-up time before collecting data for analysis.

9.3 Types of Simulations with Respect to Output Analysis

The appropriate method for analyzing simulation output depends fundamentally on the type of simulation being conducted. Simulations are classified into two broad categories: terminating and nonterminating.

1. Terminating Simulation

A terminating simulation is one in which a specific event E defines a natural stopping point for the simulation. The simulation is run from a defined initial condition until this stopping event occurs, at which point the run ends. Each replication begins under the same initial conditions and proceeds independently until the termination event.

Characteristics:

The starting state and the stopping event are both well-defined and meaningful to the problem being studied. Initial conditions play an important role, as the output is measured over the entire duration of each run, including the early transient period.

Examples:

A bank that opens at 9:00 AM and closes at 5:00 PM, with the simulation ending when the last customer has been served. A military engagement that ends when one side achieves victory. A manufacturing facility with a production target of 100 aircraft, where the simulation ends upon completion of the 100th unit. An inventory system studied over a fixed planning horizon of 120 months.

2. Nonterminating Simulation

A nonterminating simulation has no natural stopping event defined by the problem itself. The system of interest is one that operates continuously, and the goal is to study its long-run or steady-state behavior. The simulation runs for an artificially chosen length of time that is long enough for the system to reach and sustain steady-state operation.

Nonterminating simulations are further divided into three cases depending on the nature of the performance measure of interest:

Case A: Steady-State Parameter

In this case, the performance measure is a long-run average that the system approaches asymptotically. For example, in a manufacturing system, the output variable N_i represents the number of items produced in hour i. Over time, the expected production per hour stabilizes at a constant value E(N), which is the steady-state parameter of interest. The analysis must account for the initial transient period by discarding observations from the warm-up phase before estimating E(N).

Case B: Steady-State Cycle Parameter

Some systems do not reach a global steady state but do exhibit stable behavior when analyzed within a repeating cycle. For instance, a factory that operates in daily shifts with scheduled lunch breaks will have output that varies within each day, preventing a simple hourly steady state. However, the average production per day (one complete cycle) may be stable from day to day. Similarly, a call center may experience varying call volumes throughout the week, but the weekly average may be consistent. In such cases, the cycle (day, week, or shift) is used as the unit of analysis, and the steady-state cycle parameter is estimated from the average performance per cycle.

Case C: Other Parameters (No Steady State or Cycle)

In some situations, the system does not exhibit a steady state at either the observation level or the cycle level. This occurs when the system undergoes continuous structural changes, such as a manufacturing facility that changes its product mix or demand pattern every week. In such cases, there is no stable long-run distribution to estimate. The appropriate approach is to treat the simulation as terminating: define a fixed study period (e.g., three months) and analyze the output over that period, as one would in a terminating simulation.

Important Observation: Same System, Different Types

It is important to recognize that the same physical system can be studied as either a terminating or a nonterminating simulation, depending on the research question. For example, a manufacturing system studied during its initial startup phase would be treated as a terminating simulation, with the stopping event defined as the completion of the startup period. The same system studied with the goal of estimating long-run production efficiency would be treated as a nonterminating simulation. The classification depends not on the system itself, but on the objective of the study.

Summary Comparison of Simulation Types

Feature	Terminating	Nonterminating
Stopping Condition	Natural event E	Artificially chosen run length
Study Objective	Performance over a defined period	Long-run or steady-state performance
Initial Conditions	Explicitly defined and important	Less critical after warm-up
Transient Period	Included in analysis	Discarded as warm-up
Examples	Bank working day, battle, production run	Continuous factory, call center, inventory

Consolidated Exam Review

Simulation as a Statistical Experiment

Simulation is not merely a programming exercise; it is a statistical experiment conducted using a computer. As such, its output must be interpreted using the principles of probability and statistics, not taken at face value from a single run.

Why Multiple Replications Are Necessary

Because simulation output is random and subject to high variance, a single replication is insufficient to characterize the system. Multiple independent replications are required to obtain reliable estimates of performance measures.

Autocorrelation Within a Run

Observations within a single replication are not independent; each value depends statistically on its predecessors. This autocorrelation prevents the direct application of standard statistical methods and must be accounted for in the analysis.

Independence Across Replications

While observations within a run are dependent, the corresponding observations from different, independently seeded replications are statistically independent. This property is the basis for constructing valid confidence intervals.

Transient vs. Steady-State

The transient period occurs at the start of a simulation run, when the output is influenced by initial conditions and has not yet stabilized. The steady state is reached when the output distribution becomes time-invariant. In steady-state analysis, the transient (warm-up) period is discarded.

Terminating vs. Nonterminating Simulations

Terminating simulations have a natural stopping event and are analyzed over the entire defined period. Nonterminating simulations study long-run behavior and require identification and removal of the warm-up period before data collection.

Two Common Pitfalls

The two principal sources of error in output analysis are a biased estimator, where the estimated value systematically differs from the true value, and an incorrect variance estimate, which leads to misleading confidence intervals.